JAN–M–201

21 APRIL 1945

JOINT ARMY-NAVY SPECIFICATION

MERCUROUS CHLORIDE (CALOMEL)

(For Ordnance Use)

Army Number
50-11-103A

Navy Number
51M9

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army
50–11–103
1 Aug. 1941

Navy Department

A. APPLICABLE SPECIFICATIONS

A–1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition, except Small Arms Ammunition.[1]

100–2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard, Testing.

B. GRADE

B–1. This specification covers one grade of mercurous chloride known commercially as calomel.

C. MATERIAL AND WORKMANSHIP

C–1. See section E.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Moisture.*—Maximum, 0.1 percent.
E–2. *Mercurous chloride.*—Minimum, 99.6 percent.
E–3. *Nonvolatile matter.*—Maximum, 0.2 percent.
E–4. *Apparent density.*—2.2±0.2.
E–5. *Granulation.*—Mercurous chloride (calomel) shall conform to the following granulation requirements using U. S. Standard sieves in accordance with Federal Specification RR–S–366.

	Percent
Through 177-micron (No. 80), minimum	99.9
Through 74-micron (No. 200), minimum	95

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lots.*—Maximum, 5,000 pounds.

F–2. *Sampling.*—A minimum of 10 percent of the containers in the lot shall be selected by the Government inspector in such a manner as to be representative of the lot. When lots comprise less than 100 containers, either 10 containers or all containers in the lot, shall be selected. Sufficient of the material to form a primary sample of approximately ½ pound shall be removed from each selected container by means of a scoop. A 4-ounce portion of the sample shall be placed in a rubber stoppered bottle and labeled so that the container from which it was taken can be easily identified. The remaining portions of the primary samples shall be mixed together thoroughly and quartered until a composite sample of approximately 1 pound is obtained. The composite sample shall be placed in a rubber stoppered bottle and labeled to show the name of the material, manufacturer, plant, contract or order number, number of pounds in the lot, and lot number. All acceptance tests shall be made on the composite sample representative of the lot. The primary samples shall be held for possible future examination should the composite sample fail to meet the requirements of this specification.

F–3. *Inspection.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—The following tests shall be made at a Government laboratory unless otherwise specified in the contract or order:

F–4a. *Moisture.*—Transfer approximately 5 gm. of the sample to a tared weighing dish. Weigh the dish and contents accurately. Dry in a fresh sulfuric acid desiccator for 24 hours and weigh again. Calculate the loss in weight of the dish as percentage of moisture in the sample.

F–4b. *Mercurous chloride.*—Transfer an accurately weighed portion of approximately 0.7 gm. of the sample, dried as described in paragraph F–4a, to a glass-stoppered flask. Mix the sample well with 10 ml. of distilled water. Add 50 ml. of approximately 0.1 N iodine solution and 5 gm. of potassium iodide dissolved in 10 ml. of water. Stopper the flask and allow the mixture to stand, with occasional agitation, until complete solution has taken place. Titrate the residual iodine with standardized 0.1 N sodium thiosulfate solution, using 5 ml. of a freshly prepared 0.5 percent solution of soluble starch as indicator. Run a blank determination on 50 ml. of the iodine solution at the same time. Calculate the purity of the sample as follows:

$$\text{Percent mercurous chloride} = \frac{23.61(V-v)N}{W}$$

Where:

V = ml. of 0.1 N sodium thiosulfate solution used in titrating the blank

v = ml. of 0.1 N sodium thiosulfate solution used in titrating the residual iodine after solution of the sample

N = normality of the sodium thiosulfate solution

W = weight of dry sample.

F–4c. *Nonvolatile matter.*—Transfer a weighed portion of approximately 5 gm. of the sample to a tared crucible. Heat the crucible gently until no more fumes are evolved. Cool the crucible in a desiccator and

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then weigh. Calculate the increase in weight of the crucible as percentage of nonvolatile matter in the sample.

F–4d. *Apparent density.*—Tare a 100-ml. graduate cylinder, having graduations extending over approximately 7½ inches. Place on the cylinder a funnel with a stem approximately 1 inch long which is ½ inch in internal diameter. Slowly pour against the inside of the funnel 90 to 95 ml. of sample from a height of 12 inches from the base of the graduate. Do not shake or jar the cylinder. Level off the material in the graduate, read the volume occupied, and weigh the cylinder and contents. Calculate the apparent density by dividing the weight in gm. of the calomel by its volume in ml.

F–4e. *Granulation.*—Superimpose a 177-micron (No. 80) U. S. Standard sieve on a 74-micron (No. 200) U. S. Standard sieve. Transfer a weighed 50-gm. portion of the sample to the top sieve. Moisten the sample with 95 percent alcohol. Wash the particles through the sieves with a stream of water, breaking up all lumps by gentle brushing with a camel's hair brush. Dry each sieve and the residue remaining on it at approximately 100° C. Brush the dried residue remaining on each sieve into a tared weighing dish and weigh. From the weights of the residues remaining on the sieves, calculate the percentage of material passing through each sieve.

F–5. *Retests.*—If the composite sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having an analysis of each primary sample made at his own expense. The manufacturer may then remove or replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and submit the lot for acceptance, provided that the markings on the container are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, calomel shall be packed in moistureproof packages having a maximum capacity of 25 pounds.

G–2. *Packing.*—Unless otherwise specified calomel shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

G–3. *Marking.*—Unless otherwise specified, each container shall be plainly marked with the following information completed:

MATERIAL _____
SPECIFICATION NO. _____
STOCK NO. _____
CONTRACTOR _____
MANUFACTURER _____
CONTRACT NO. _____
GROSS WEIGHT _____
DATE _____
LOT NO. _____

In addition to the foregoing, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook.

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H. NOTES

H–1. Requests, requisitions, schedules, and contracts or orders should contain the title of the specification, the number and date.

H–2. *Use.*—Calomel covered by this specification is intended for use in the manufacture of tracer composition.

H–3. Copies of Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

H–4. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases) and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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